Binary-Model Applications and Final Review

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ECO 351 – Introduction to Econometrics

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Recall: the probability comparison

Last class, probit predicted denial probabilities 0.0962, 0.1570, and 0.2388 at PIR 0.30, 0.40, and 0.50.

A student reports: "A 0.10 increase in PIR always raises denial probability by 6.08 percentage points."

Use one of the other comparisons to assess that statement. What does the probit coefficient itself change?

Debrief: the starting value matters

From PIR 0.40 to 0.50, the probability rises by

$$0.2388 - 0.1570 = 0.0818,$$

or about 8.18 percentage points.

The coefficient changes the index by a constant amount per unit PIR. The normal CDF converts that index change into a starting-value-dependent probability difference.

We now ask how software chooses the coefficients and quantifies the uncertainty of a difference.

The probability of an observed outcome

For an observed denial, a model assigning high denial probability makes that observation more likely. For an approval, a model assigning low denial probability makes it more likely.

$$L_i = p_i^{Y_i}(1 - p_i)^{1-Y_i}$$

A likelihood for several applicants

Constructed independent applicants; a proposed model supplies each denial probability p_i.

Applicant	Observed Y	p_i	Observed-outcome probability
A	1	.7	.7
B	0	.2	$1 - .2 = .8$
C	1	.6	.6

$$L = .7 \times .8 \times .6 = .336$$

Exercise: comparing two proposed models

Two applicants have outcomes (denied, approved). Model A predicts denial probabilities (0.8,0.3); Model B predicts (0.6,0.6). Under independence, which assigns the observed pair a higher likelihood?

Debrief: the observed approval also matters

$$L_A = 0.8(1 - 0.3) = 0.56, \qquad L_B = 0.6(1 - 0.6) = 0.24$$

A assigns higher probability to this observed pair. The second contribution uses the probability of approval, not denial.

Maximum likelihood chooses the model coefficients that maximize the likelihood of the observed sample. Software typically works with its logarithm.

Stata: retaining predictions for a contrast

```
. margins, at(pir=(0.30 0.40 0.50)) post
Adjusted predictions                                      Number of obs = 2,380
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   .0961532   .0069017    13.93   0.000       .082626    .1096803
          2  |   .1569683   .0101166    15.52   0.000      .1371402    .1767965
          3  |   .2387884    .024485     9.75   0.000      .1907987    .2867781
------------------------------------------------------------------------------
```

The `post` option retains these estimates for the following `lincom` commands.

Stata: uncertainty for a finite difference

```
. lincom _b[2._at] - _b[1._at]
 ( 1)  - 1bn._at + 2._at = 0
------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0608152   .0099471     6.11   0.000     .0413193    .0803111
------------------------------------------------------------------------------
```

The 0.30-to-0.40 change is about 6.08 percentage points, with a 95% interval of about $[4.13, 8.03]$ percentage points.

The contrast uses uncertainty in both predictions and their covariance.

Exercise: another probability difference

```
. lincom _b[3._at] - _b[2._at]
 ( 1)  - 2._at + 3._at = 0
------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0818201   .0163231     5.01   0.000     .0498274    .1138127
------------------------------------------------------------------------------
```

1. Which PIR change does this contrast represent?
2. Report its point estimate and interval in percentage points.
3. Is this a predicted probability for a single applicant?

Debrief: a larger fitted change

The contrast compares PIR 0.40 with 0.50:

$$100\Delta\widehat{p} = 8.18 \text{ percentage points},$$

with a 95% interval of approximately $[4.98, 11.38]$ percentage points.

Both estimates come from the same probit model. Different starting values generate different finite changes.

Logit: the comparable prediction contrast

```
. lincom _b[2._at] - _b[1._at]
 ( 1)  - 1bn._at + 2._at = 0
------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0635805   .0111931     5.68   0.000     .0416424    .0855186
------------------------------------------------------------------------------
```

This follows logit in the do-file. For PIR 0.30 to 0.40, logit gives about 6.36 percentage points, compared with probit's 6.08.

Comparing probabilities makes more sense than comparing the raw index coefficients.

Average marginal effects and indicator changes

For a continuous regressor, an **average marginal effect (AME)** averages the fitted local probability slope across the observed applicants.

For an indicator, compare each applicant's fitted probability with the indicator set to 1 versus 0, keeping their other regressors fixed. Then average those discrete differences.

`margins, dydx(...)` handles this distinction when the model uses `i.` for the indicator.

Neither average is automatically a causal effect.

Stata: average effects in a model with an indicator

```
. margins, dydx(pir black)
Average marginal effects                                  Number of obs = 2,380
------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   .5182777   .0905715     5.72   0.000     .3407609    .6957945
     1.black |   .1664023   .0240036     6.93   0.000     .1193561    .2134485
------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

This follows `logit deny c.pir i.black, vce(robust)`. The `1.black` row gives an average adjusted probability difference of about 16.64 percentage points.

This conditional association does not by itself establish a causal discrimination effect.

Exercise: a report that overstates the evidence

A report states:

> *The logit coefficient is larger than the probit coefficient, so logit finds a much larger causal effect. Its bounded predictions eliminate omitted-variable bias.*

Rewrite this using the common 0.30-to-0.40 comparison. Identify the two separate mistakes.

Debrief: comparable magnitudes and causal limits

For the same PIR change and sample, the fitted denial-probability increases are about 6.36 percentage points under logit and 6.08 under probit.

Both models constrain predicted probabilities. Neither constraint establishes that omitted applicant characteristics are unrelated to PIR.

Report the conditioning variables, uncertainty, and limitations alongside the fitted comparison.

Cumulative review: the source of the comparison

Match each research concern to the relevant idea, then state its limitation.

Concern	Candidate approach
A relationship bends as income rises	Polynomial or logarithmic form
Persistent city differences confound a fare comparison	City fixed effects
Fares respond to unobserved demand shocks	An external fare instrument
The outcome is a zero-one purchase decision	LPM, probit, or logit

Which approaches change functional form, and which need a new identifying argument?

Exercise: choosing an empirical strategy

A city panel records bus ridership, fares, and income. Cities raise fares when unobserved demand is strong. An external operating-cost change might predict fares, but could also reduce service frequency.

1. What can city fixed effects remove?
2. Why can the fare regressor remain endogenous?
3. What threatens using the cost change as an instrument?

Debrief: matching methods to threats

City effects can absorb fixed differences such as persistent urban layout. They leave changing city-specific demand shocks unresolved.

Those shocks can influence both fares and ridership. An operating-cost instrument needs a credible argument that any ridership response runs through the modeled fare channel.

Changes in service frequency can undermine that argument. Logging fares or reporting clustered standard errors does not remove the channel.

Preparing to discuss a project

For the project-sharing session on Monday, 23 November, prepare to explain:

- Your question, data, and the comparison behind your main coefficient.
- One informative figure and the main estimate in meaningful units.
- The strongest justified conclusion and the most important remaining limitation.

Bring a question that you would like the class to discuss.